

Computational Social Science with Images and Audio

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Human vision is interesting (not only for social scientists).

- ▶ Vision is an incredibly powerful sense
- ▶ Allows us to interact with the physical world without making any physical contact
- ▶ A *substantial* part of our brain is somehow involved in visual perception
 - ▶ Of course, there are limitations of human vision (e.g., low quantification skills)
- ▶ Human vision is interesting on many levels:
 - ▶ Fundamental: How do biological vision systems operate?
→ Not part of this class (or only indirectly)
 - ▶ Aggregate/societal: In what ways does what and how we see shape societal outcomes?
→ The focus of this class

For which of these parties would you vote?



For which of these parties would you vote?



Figure: Fictitious party logos created with ChatGPT

Computer vision comes with many definitions.

- ▶ David Marr: automating human visual processes
- ▶ Others see it as an information-processing task
- ▶ Berthold Horn: inverting image formation
- ▶ So, perhaps: the inverse of graphics?
- ▶ In this class, we look at techniques to analyze visual data (images, videos) in large quantities to study social science questions
 - ▶ We start where the visual data already exists: we do not cover, from a technical viewpoint, how to get that visual data (e.g., optics)
- ▶ Furthermore, we also discuss the societal consequences of computer vision applications
 - ▶ CV is both a tool and a research subject

Essentially, our class starts with something like this.

```
array([[131, 131, 131, ..., 107, 106, 106],
       [131, 132, 132, ..., 107, 107, 106],
       [132, 132, 133, ..., 108, 107, 106],
       ...,
       [192, 192, 192, ..., 189, 187, 185],
       [193, 193, 193, ..., 189, 187, 185],
       [194, 194, 194, ..., 188, 186, 184]], dtype=uint8)
```

Figure: An array of (549, 976)

What the array represents:

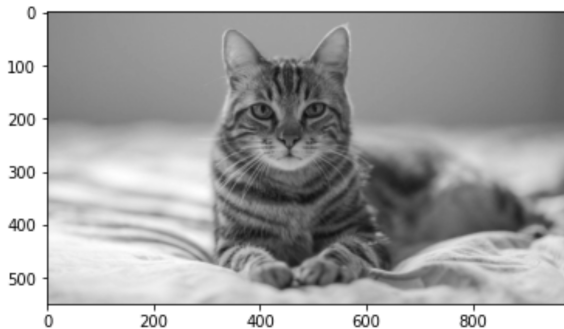


Figure: Meow.

What about the social science part?

- ▶ Social science is the study of the complex interplay between individuals, groups, and institutions in society and ...
- ▶ How individuals, groups, and institutions shape and are shaped by social, cultural, political, and economic forces
- ▶ Auguste Comte: a discipline that studies society positively/scientifically
- ▶ Max Weber: the study of social action, or human behavior as it relates to the social context in which it occurs

Computer vision is increasingly important in social science.

- ▶ Images significantly contribute to and mirror the social/political landscape
- ▶ Digitization has led to an increased accessibility of such images in everyday life
 - ▶ Consequently, this provides new and valuable opportunities for researchers and practitioners alike (both in CV and social science)
- ▶ Advances in computer vision significantly ease the process of using images as data
 - ▶ For example, deep learning techniques for identifying objects, measuring visual sentiment, or recognizing faces
 - ▶ Innovation in computer vision has large societal effects!
- ▶ Great opportunity comes with great responsibility...

Digitization has led to an explosion in visual material.

- ▶ There has been an explosion in the production and consumption of images
- ▶ On Youtube, an estimated 3.7 million videos are uploaded per day¹
- ▶ Users upload 1.3 billion images to Instagram per day²
- ▶ (Such figures are estimates and can change fast but are enormous in any case...)
- ▶ Can you think of other major sources of visual data?
- ▶ Can you think of ways how the image collection would influence downstream work with those images?

¹Adavelli, M., & Tonogbanua, L. (2023). How Many Videos Are Uploaded to YouTube A Day? TechJury. <https://techjury.net/blog/how-many-videos-are-uploaded-to-youtube-a-day/>

²Growcoot, M. (2023). Almost All Photos Are Now Taken on Smartphones, According to Study. PetaPixel. <https://petapixel.com/2023/06/20/almost-all-photos-are-now-taken-on-smartphones-according-to-study/>

Studying images in social and political life is not new.

- ▶ For instance, research has shown that images are key to agenda-setting and framing in newspaper coverage
- ▶ Images can impact people's perceptions of political candidates and their votes and ...
- ▶ Encourage (or discourage) political participation
- ▶ Generally, images seem to have a greater effect than written and spoken content in capturing attention
- ▶ They can ease information processing, improve information recall, and evoke emotions³

³Examples of relevant papers can be found in Webb Williams, N., Casas, A., and Wilkerson, J. (2020). Images as Data for Social Science Research: An Introduction to Convolutional Neural Nets for Image Classification (Elements in Quantitative and Computational Methods for the Social Sciences). Cambridge: Cambridge University Press. doi:10.1017/9781108860741

This is an early example of how visual material in the media shapes political discourse.

- ▶ A “classic”: Gitlin’s “The Whole World Is Watching: Mass Media in the Making and Unmaking of the New Left” (1980)

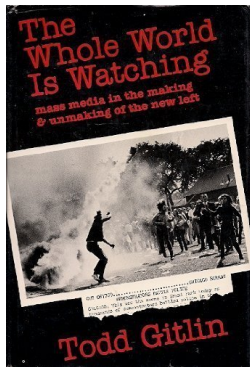


Figure: Gitlin (1980)

The abundance of images, however, is new!

- ▶ We can challenge existing knowledge in new ways, and generate new knowledge
- ▶ Social science work with visual data comes in two categories:
 - ▶ Images measure variables of interest in new ways
 - ▶ Images are the object of study (in a causal or non-causal framework, with image-based dependent or independent variables)
 - ▶ (Fuzzy, overlapping distinction)
- ▶ Can you think of examples for the above categories?
 - ▶ Can we detect fraud in elections by using original image databases of the vote-tally sheets to detect alterations?⁴
 - ▶ What makes politicians' Instagram posts popular?⁵

⁴Cantu, F. (2019). The Fingerprints of Fraud: Evidence from Mexico's 1988 Presidential Election. *American Political Science Review*, 113(3), 710-726.

⁵Peng, Y. (2021). What Makes Politicians' Instagram Posts Popular? Analyzing Social Media Strategies of Candidates and Office Holders with Computer Vision. *The International Journal of Press/Politics*, 26(1), 143-166.

Can we detect election fraud with images? (Cantu, 2019)

The figure displays four examples of Mexican election tally sheets, labeled A, B, C, and D. Each sheet is a grid with columns for 'VOTACIONES HECHAS EN LA CUBA' (Votes cast in the booth), 'VOTOS REGISTRADOS EN LA CUBA' (Votes registered in the booth), and 'VOTOS CONTABILIZADOS EN LA CUBA' (Votes tallied in the booth). The sheets show various handwritten numbers in different colors (blue, green, black) and positions, indicating potential alterations or fraud.

A

VOTACIONES HECHAS EN LA CUBA	VOTOS REGISTRADOS EN LA CUBA	VOTOS CONTABILIZADOS EN LA CUBA
131	131	
02	7	
128	138	
00		
128	138	

B

VOTACIONES HECHAS EN LA CUBA	VOTOS REGISTRADOS EN LA CUBA	VOTOS CONTABILIZADOS EN LA CUBA
19		
120		
101		
1		
10		
37		
1		
22		
2		
273		
14		
277		

C

VOTACIONES HECHAS EN LA CUBA	VOTOS REGISTRADOS EN LA CUBA	VOTOS CONTABILIZADOS EN LA CUBA
12		
1304		
20		
1		
2		
3		
1432		
1		
1432		

D

VOTACIONES HECHAS EN LA CUBA	VOTOS REGISTRADOS EN LA CUBA	VOTOS CONTABILIZADOS EN LA CUBA
309	309	
22	22	
301	301	
301	301	

Figure: Examples of altered vote tallies in Mexico, 1988 (from Cantu, 2019)

Politicians' Instagram: what is popular? (Peng, 2021)



Figure: Screenshot of New Jersey Senator Bob Menendez' Instagram (2017)

Taking a step back: What is an image?

- ▶ Simplest form (single-channel): two-dimensional function $f(x, y)$
 - ▶ Maps a coordinate pair to an integer/real value related to the intensity/color of the point
 - ▶ Single-channel: binary or mono-chrome, gray-scale, black/white images
- ▶ Each point is called a **pixel** (picture element) or pel
- ▶ Multiple channels possible, such as RGB:
 - ▶ Color represented using three channels
 - ▶ Pixel at point (x, y) represented as $(r_{x,y}, g_{x,y}, b_{x,y})$
- ▶ Channel-specific pixel value represented as an integer between 0 and 255 or a floating-point value in $[0, 1]$

Taking a step back: What is an image?

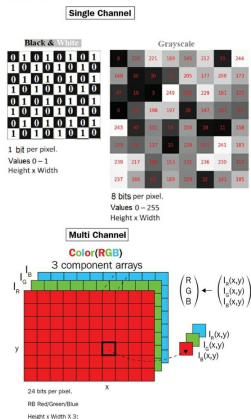


Figure: Dey (2018)⁶

⁶Dey, S. (2018). Hands-On Image Processing with Python: Expert Techniques for Advanced Image analysis and Effective Interpretation of Image Data. Packt Publishing Ltd.

RGB (0-255) cannot represent all possible colors!

- ▶ Standard RGB (0-255) is called **sRGB**: $256 \times 256 \times 256 = 16.7$ million colors
- ▶ Human vision is more sensitive: can distinguish an estimated 10-100 million colors
- ▶ Example: Many vibrant cyan-green colors in tropical waters cannot be represented in sRGB
 - ▶ They fall outside the sRGB “color gamut” (i.e., the representable colors)
- ▶ Other color spaces use different approaches but with important distinctions:
 - ▶ **Color gamut** (range/variety) $\neq$ **Color depth** (number of discrete values)

Different color spaces: gamut vs. depth

- ▶ **Adobe RGB:** Wider gamut, still 0-255 but different color mapping
 - ▶ Same total number of colors: $256 \times 256 \times 256 = 16.7$ million
 - ▶ $\text{RGB}(255,0,0)$ in sRGB $\neq$ $\text{RGB}(255,0,0)$ in Adobe RGB
 - ▶ “Stretches” the 0-255 range to cover wider area of possible colors
- ▶ **ProPhoto RGB:** Even wider gamut, can use 16-bit values (0-65,535) per channel
 - ▶ 8-bit ProPhoto: Same 16.7 million colors, just different selection
 - ▶ 16-bit ProPhoto: $65,536^3 = 281$ trillion colors
- ▶ In practice, this limitation doesn't matter much:
 - ▶ Most consumer monitors can only display 70-99% of sRGB anyway
 - ▶ sRGB remains the web and digital photography standard
 - ▶ None of this matters for our purposes, but it is still interesting ;)

CNNs have been the CV workhorse for >10 years.

(This may end soon, or has already ended; see below...)

- ▶ Convolutional neural nets (abbreviated as CNNs or CovNets) are specialized neural networks for CV tasks
 - ▶ For various tasks, such as object detection, facial recognition, image segmentation, sentiment analysis
- ▶ Much of the recent image-based social science research uses CNN
- ▶ CNNs are not the only horse in the stable:
 - ▶ Classic techniques: in the pre-deep learning and pre-CNN era, CV relied on hand-crafted features and algorithms
 - ▶ Edge detection, color histograms, template matching, ...
 - ▶ Other neural nets: recurrent neural networks (RNN), generative adversarial networks (GAN), transformers, stable diffusion models
 - ▶ RNN are often used for video analysis, GAN for image generation
 - ▶ Hybrid approaches: traditional approaches and deep learning
 - ▶ Unsupervised and semi-supervised learning: e.g., clustering

Flashback: How CNNs became the CV workhorse.

- ▶ Before 2012: Computer vision relied on hand-crafted features
- ▶ September 2012: AlexNet won the ImageNet Challenge
 - ▶ ImageNet Challenge: annual competition to classify images into 1,000 categories from a dataset of millions of labeled images
 - ▶ Top-5 error rate: 15.3% vs. runner-up's 26.2%
 - ▶ 1st time a deep CNN outperformed traditional methods large-scale
- ▶ AlexNet's key innovations:
 - ▶ Used GPUs for training (revolutionary at the time)
 - ▶ ReLU activation functions instead of sigmoid/tanh
 - ▶ Dropout regularization to prevent overfitting
 - ▶ Data augmentation (horizontal reflections, color alterations)
- ▶ Demonstrated CNNs could automatically learn features from data
- ▶ Launched the deep learning revolution in CV (2012-2020s)

The AlexNet revolution is also seen in the numbers.

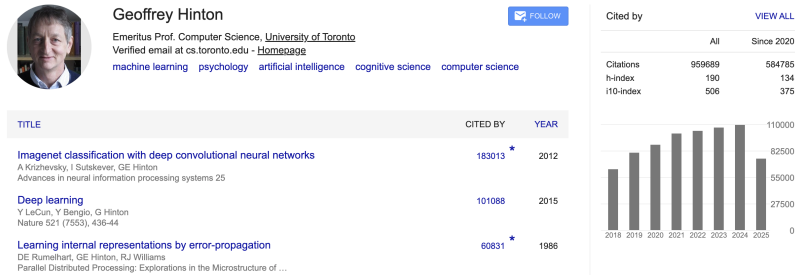


Figure: Geoffrey Hinton's Google Scholar profile (as of September 2025)

Before that, classic techniques required explicit “rules” .

- ▶ Imagine you search for images with a specific political leader
- ▶ Should you create a checklist of physical characteristics?
 - ▶ Shape of their nose, size of forehead?
 - ▶ Not robust (facing the camera? sunglasses?)
 - ▶ An infinite list of rules?
- ▶ Fairly general features can be helpful for various tasks
 - ▶ Corners
 - ▶ Blobs (dark/bright region)
- ▶ **Feature detection:** identification of points of interest
- ▶ Explicit feature detection in classic or “traditional” machine learning

Classic techniques typically require more explicit “rules”.

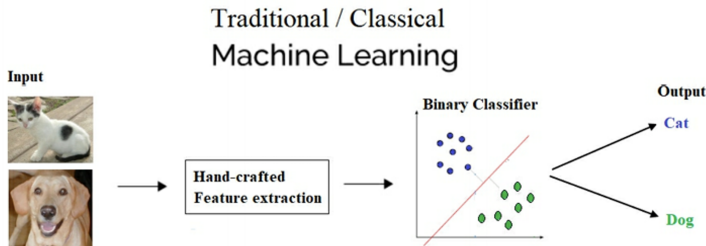


Figure: Dey (2018)⁷

⁷ Dey, S. (2018). Hands-On Image Processing with Python: Expert Techniques for Advanced Image analysis and Effective Interpretation of Image Data. Packt Publishing Ltd.

Neural nets provide a more end-to-end approach.

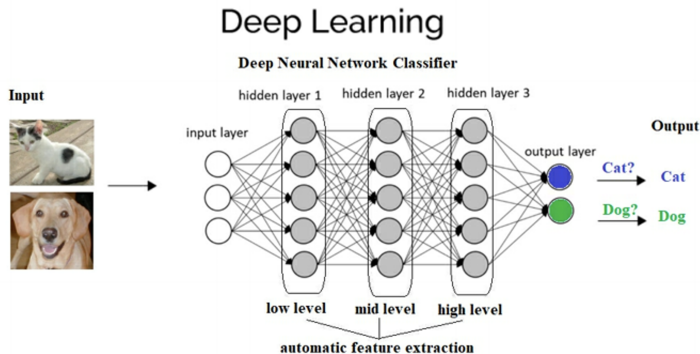


Figure: Dey (2018)⁸

⁸Dey, S. (2018). Hands-On Image Processing with Python: Expert Techniques for Advanced Image analysis and Effective Interpretation of Image Data. Packt Publishing Ltd.

CNNs contain input layer + hidden layers + output layer.

- ▶ Typically: input layer $\rightarrow$ hidden layers $\rightarrow$ output layer
- ▶ What do different convolutional layers do?
 - ▶ Layers close to the input layer learn low-level features (e.g., lines)
 - ▶ Middle layers learn complex abstract features (combining lower-level features)
 - ▶ Layers closer to the output interpret the extracted features in the light of the classification task

The ViT breakthrough: CNNs challenged (2021)

- ▶ “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale”
- ▶ Vision Transformer (ViT) showed pure transformers could match/beat CNNs on ImageNet
 - ▶ Achieved 88.55% accuracy on ImageNet while requiring 4x fewer computational resources than comparable CNNs
 - ▶ 1st successful application of pure transformer architecture to CV
- ▶ ViT's key innovation: treat images as sequences of patches
 - ▶ Split image into 16×16 pixel patches (hence “16x16 Words”)
 - ▶ Each patch becomes a “token” like words in NLP
 - ▶ No convolutions needed!
- ▶ Demonstrated that transformers could capture global image context
 - ▶ CNNs build features hierarchically (local $\rightarrow$ global)
 - ▶ ViTs can model global relationships from the very first layer

ViT treats pixel patches like “tokens”

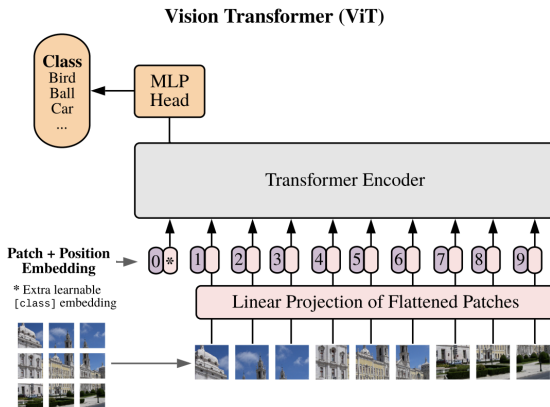


Figure: Dosovitskiy et al. (2021)⁹

⁹Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. International Conference on Learning Representations (ICLR).

From visual models to image-text models

- ▶ CNNs and ViTs turn images into visual representations.
- ▶ Many task-specific classifiers connect those representations to a fixed set of labels
- ▶ Image-text models learn representations that connect images with language
 - ▶ Often trained using paired images and captions
- ▶ Researchers can then describe candidate concepts in natural language
- ▶ CLIP is a prominent example

CLIP aligns images and text

- ▶ CLIP has an image encoder and a text encoder
 - ▶ Both map their input into a shared embedding space
- ▶ During training, matching image-text pairs are brought closer together in that space
- ▶ Both representations occupy the same learned semantic geometry
- ▶ To classify an image, compare it with text descriptions of possible categories.¹⁰
 - ▶ The category with the highest similarity score is the model's prediction

¹⁰Radford, A., et al. (2021). Learning Transferable Visual Models From Natural Language Supervision. ICML, 139, 8748–8763.

Describing Images with Text Concepts

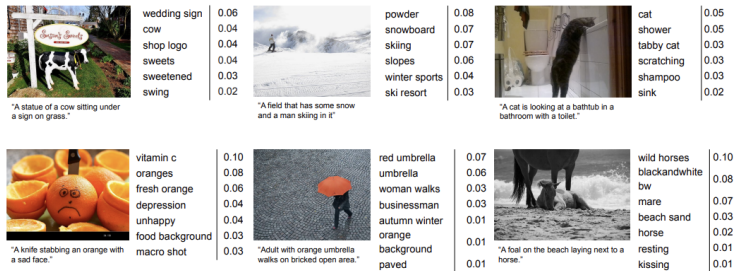


Figure 2: Example images from MSCOCO shown with their captions below and their concept decompositions on the right. We display the top seven concepts for visualization purposes, but images in the figure had decompositions with 7-20 concepts.

How does a vision-language model work?

- ▶ A common design combines three components:
 - ▶ A vision encoder turns the image into visual features or tokens.
 - ▶ A connector passes those features to the language model.
 - ▶ The language model uses the visual information and the text prompt to generate a response
- ▶ The response is generated one token at a time.
- ▶ Architectures differ: some use a connector, while others use cross-attention to link visual and language features¹¹

¹¹See Liu, H., et al. (2023). Visual Instruction Tuning. NeurIPS; and Alayrac, J.-B., et al. (2022). Flamingo: A Visual Language Model for Few-Shot Learning. NeurIPS.

Using vision-language models for social science

- ▶ Ask the same theory-guided questions about a collection of images
- ▶ For example, code protest coverage for demonstrators, police, banners, or confrontations
- ▶ Turn model responses into candidate measures of image content or visual framing
- ▶ Treat those responses as measurements that require validation
- ▶ We will discuss model training and validation in more detail in the following lectures